

Unit I – Basics, Zeroth and First Law

Detailed lecture notes — systems and properties, work and heat, the Zeroth law and temperature, the first law for closed and open systems, with solved numericals

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SYLLABUS

UNIT I — BASICS, ZEROTH AND FIRST LAW (9 + 3 periods)

Review of Basics – Thermodynamic systems, Properties and processes, Thermodynamic Equilibrium – Displacement work – P-V diagram. Thermal equilibrium – Zeroth law – Concept of temperature and Temperature Scales. First law – application to closed and open systems – steady and unsteady flow processes.

Course Outcome CO1: Apply the first law of thermodynamics to closed and open systems for thermal equipment.

Text: P. K. Nag, *Engineering Thermodynamics*, 6th Edn., McGraw-Hill (Chapters 1–5).

References: Cengel & Boles, *Thermodynamics – An Engineering Approach*, 8th Edn.; R. K. Rajput, *A Text Book of Engineering Thermodynamics*, 5th Edn.

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Learning objectives. After studying this unit you should be able to: (i) define a thermodynamic system, its boundary and surroundings and classify systems as closed, open or isolated; (ii) distinguish intensive from extensive properties, point functions from path functions and processes from cycles; (iii) state the conditions of thermodynamic equilibrium and explain the quasi-static and reversible process; (iv) state the Zeroth law and relate the Celsius, Kelvin, Fahrenheit and Rankine scales; (v) evaluate displacement work for constant-pressure, constant-volume, isothermal, polytropic and adiabatic processes from the p – V diagram; (vi) state and apply the first law to cycles and to non-flow processes, including tabulated multi-process cycles; (vii) derive the steady flow energy equation and apply it to nozzles, diffusers, turbines, compressors, heat exchangers and throttling devices; (viii) set up energy balances for unsteady (charging) processes.

1. Introduction – what thermodynamics is about

Thermodynamics is the science of **energy** and its transformations. Every engineering device that produces power, provides refrigeration or processes a fluid – a steam turbine, a petrol engine, a refrigerator, a gas compressor, a boiler – is governed by two great principles: the **first law**, which is the law of conservation of energy, and the **second law**, which tells us in which direction processes can proceed and how much of the available energy can be converted into useful work. Unit I lays the foundation: the vocabulary of systems and properties, the Zeroth law that gives us temperature, and the first law applied to closed and open systems.

The name comes from the Greek *therme* (heat) and *dynamis* (power). Historically the subject grew out of the effort, in the nineteenth century, to understand how much work could be obtained from the heat of burning coal in a steam engine. Today it underpins the design of power plants, internal combustion engines, gas turbines, refrigeration and air-conditioning systems, chemical process plants and even biological and cosmological models.

1.1 Macroscopic and microscopic points of view

There are two ways of describing the behaviour of matter.

- **Macroscopic (classical) approach.** The system is described by a small number of measurable bulk properties – pressure, volume, temperature – without any reference to the structure of matter. Only a few variables are needed and they can be measured directly with instruments. This is the approach of classical thermodynamics and of this course.
- **Microscopic (statistical) approach.** Matter is treated as a collection of a very large number of molecules, each with its own position and velocity; the bulk behaviour is obtained by statistical averaging of molecular behaviour. This is the approach of statistical thermodynamics and kinetic theory.

The two approaches give the same results for the quantities that can be measured; the macroscopic approach is simpler, and its results are independent of any assumption about molecular structure.

1.2 The concept of continuum

In the macroscopic approach the substance is regarded as a **continuum** – a continuous distribution of matter with no voids – so that properties such as density and pressure can be defined at every point as limits of the ratio of mass (or force) to a small volume (or area). The assumption is valid as long as the smallest volume of interest contains a very large number of molecules, i.e. when the mean free path of the molecules is negligible compared with the dimensions of the system. It fails only for highly rarefied gases (very high vacuum, upper atmosphere) where the molecules are too few for averaging to be meaningful.

1.3 Where thermodynamics is used

- **Power generation:** steam and gas turbine plants convert the chemical energy of fuel into electricity; the efficiency of every plant is limited by thermodynamic laws.
- **Internal combustion engines:** the Otto and Diesel cycles, mean effective pressure and fuel economy are first- and second-law calculations.
- **Refrigeration and air-conditioning:** pumping heat from a cold space to warm surroundings, within the limits set by the second law.
- **Flow machinery:** turbines, compressors, pumps, nozzles, boilers and condensers are analysed with the steady flow energy equation derived in this unit.

2. Thermodynamic systems, boundary and surroundings

Every thermodynamic analysis begins by choosing the **system** – the definite quantity of matter or the region in space selected for study. Everything external to the system is the **surroundings**. The **boundary** is the real or imaginary surface, of zero thickness, that separates the system from its surroundings; it may be fixed (the walls of a rigid tank) or movable (the face of a piston), and energy – and in some cases mass – crosses it. The system together with its surroundings is called the **universe**.

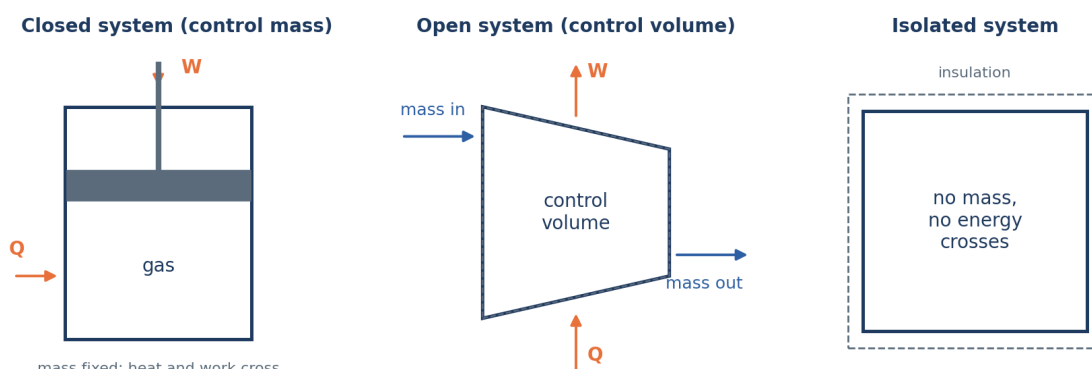


Figure 1.1 The three types of thermodynamic system: closed, open (control volume) and isolated.

2.1 Types of systems

Type	Mass crosses boundary?	Energy crosses boundary?	Also called	Examples
Closed system	No	Yes – heat and/or work	Control mass	Gas in a piston–cylinder; a sealed pressure cooker; a car battery
Open system	Yes	Yes	Control volume	Turbine, compressor, boiler, nozzle, pump, water heater
Isolated system	No	No	—	Ideal insulated rigid thermos flask; the universe (system + surroundings)

A **closed system** has a fixed mass; the boundary may move (the gas in a cylinder expands and pushes a piston), but no matter enters or leaves. An **open system** is best analysed as a **control volume** – a fixed region in space, enclosed by a **control surface**, through which mass flows carrying its energy in and out; almost all practical machines are open systems. An **isolated system** exchanges neither mass nor energy with anything outside it, so its total energy is constant; it is an idealisation, but a well-insulated rigid sealed vessel approaches it, and the combination of a system with all of its surroundings is by definition isolated.

KEY POINT Always draw the system and mark its boundary before writing any equation. The boundary decides which heat, work and mass transfers enter the analysis – the same physical situation can be a closed or an open system depending on where the boundary is drawn.

2.2 Homogeneous and heterogeneous systems

A **phase** is a quantity of matter that is homogeneous throughout in chemical composition and physical structure (solid, liquid or gas). A system consisting of a single phase is **homogeneous** (air in a room, water in a tank); a system consisting of two or more phases is **heterogeneous** (water and steam in a boiler drum, ice and water in a glass).

3. Properties, state, process, path and cycle

3.1 Properties

A **property** is any measurable characteristic of a system whose value depends only on the state of the system and not on how the system reached that state: pressure p , temperature T , volume V , mass m , density ρ , internal energy U , enthalpy H , entropy S , and so on. Properties are classified as follows.

Basis	Intensive property	Extensive property
Dependence on mass	Independent of the mass of the system	Proportional to the mass of the system
If the system is divided into two halves	Value unchanged in each half	Value halved in each half
Examples	Pressure, temperature, density, specific volume, specific internal energy, specific enthalpy	Volume, mass, total internal energy, enthalpy, entropy
Notation	Lower-case letters (except p and T)	Capital letters

Any extensive property divided by the mass becomes a **specific** property, which is intensive: specific volume $v = V/m$ (m^3/kg), specific internal energy $u = U/m$ (kJ/kg), specific enthalpy $h = H/m$ (kJ/kg). The quick test: imagine cutting the system in half – if the value changes, the property is extensive.

3.2 State

The **state** of a system is its complete condition, identified by the values of its properties. When the properties have definite values the system is said to be in a definite state; on a property diagram (p – V , T – s) a state is a single point. For a simple compressible system (a pure substance in the absence of motion, gravity, electricity, magnetism and surface effects) any **two independent intensive properties** fix the state – the *two-property rule*; all other properties can then be found.

3.3 Process, path and cycle

- **Process:** the change of a system from one equilibrium state to another. Processes are named after the property that is held constant: *isobaric* (constant pressure), *isochoric* (constant volume), *isothermal* (constant temperature), *adiabatic* (no heat transfer), *isentropic* (constant entropy).
- **Path:** the succession of states through which the system passes during the process – the curve joining the end points on a property diagram.
- **Cycle:** a sequence of processes that begins and ends at the same state. Since the initial and final states are identical, the net change of every property over a cycle is zero: $\oint dp = 0$, $\oint dV = 0$, $\oint dU = 0$. A process has definite, generally different, end states; a cycle returns to its starting point.

3.4 Point functions and path functions

Feature	Point function	Path function
Depends on	The end states only	The particular path followed between the states

Feature	Point function	Path function
Differential	Exact: dp, dV, dT, dU (integrates to $p_2 - p_1$, etc.)	Inexact: $\delta Q, \delta W$ (integrates to Q_{1-2}, W_{1-2} , never to " $Q_2 - Q_1$ ")
Cyclic integral	$\oint d(\text{property}) = 0$	$\oint \delta Q$ and $\oint \delta W$ are not zero in general
Examples	All properties: p, V, T, U, H, S	Heat Q and work W

KEY POINT Heat and work are *not* properties. A system *possesses* energy; it never "contains" heat or work – these exist only while they are crossing the boundary. That is why we write $\Delta U = U_2 - U_1$ but never ΔQ ; the heat transferred in a process is written Q_{1-2} or simply Q .

4. Thermodynamic equilibrium and the quasi-static process

4.1 Thermodynamic equilibrium

A system is in **thermodynamic equilibrium** when its properties do not change with time and it has no tendency to change when isolated from its surroundings. Three conditions must be satisfied simultaneously:

- **Mechanical equilibrium** – there are no unbalanced forces within the system or between the system and its surroundings; the pressure is uniform (allowing for gravity) and no movable part is being accelerated.
- **Thermal equilibrium** – the temperature is uniform throughout the system and equal to that of the surroundings in contact with it, so that there is no heat flow.
- **Chemical equilibrium** – there is no chemical reaction and no transfer of matter from one part of the system to another (no diffusion, no phase change in progress).

Classical thermodynamics deals only with equilibrium states; the properties of a system can be defined only when it is in equilibrium, because a non-uniform system has no single value of pressure or temperature.

4.2 Quasi-static (quasi-equilibrium) process

If a gas in a cylinder is compressed suddenly, the pressure near the piston rises before the rest of the gas is affected, the gas is not in equilibrium during the process, and its state cannot be plotted. If instead the piston is moved extremely slowly – for example by removing tiny weights one at a time – the departure from equilibrium at every instant is infinitesimal and the system passes through a continuous succession of equilibrium states. Such an infinitely slow process is called a **quasi-static** or **quasi-equilibrium** process.

Its practical significance is twofold: (i) every intermediate state is a defined equilibrium state, so the process can be drawn as a continuous curve on a property diagram; and (ii) the work done can be evaluated as $\int p dV$ using the system pressure, which is the basis of all the work expressions derived in this unit. A quasi-static process is an idealisation that real processes approach when the change is slow compared with the time the system needs to reach equilibrium (for gases this relaxation time is very short, so many real processes can be treated as quasi-static).

5. Reversible and irreversible processes

A **reversible process** is one which, having taken place, can be reversed so that both the system and its surroundings are restored to their initial states, leaving no trace anywhere. During a reversible process the system passes through equilibrium states, and the work and heat transfers of the forward and reverse

processes are equal and opposite. A reversible process is therefore quasi-static *and* free of dissipative effects.

An **irreversible process** cannot be reversed without leaving some change in the surroundings. All natural processes are irreversible; the reversible process is a limiting ideal against which real processes are judged. The main causes of irreversibility are:

- **Friction** – solid friction, fluid friction (viscosity) and the flow of electricity through a resistance, which convert work into heat that cannot be fully recovered.
- **Unrestrained (free) expansion** – a gas expanding into a vacuum does no work; compressing it back requires work from the surroundings.
- **Heat transfer across a finite temperature difference** – the heat cannot flow back from the colder body to the hotter body without external work.
- **Mixing of two different substances** or of the same substance at different states; spontaneous chemical reactions; inelastic deformation; magnetic hysteresis.

Free expansion. A rigid insulated vessel is divided by a partition; one side contains a gas and the other is evacuated. When the partition is removed the gas rushes into the vacuum. No work is done (there is no resisting force at the moving boundary and no piston is displaced), no heat is transferred (insulated), so by the first law the internal energy is unchanged; for an ideal gas the temperature is unchanged. The process is irreversible because the gas will never spontaneously collect itself back into one half of the vessel, and restoring it requires compression work from the surroundings. Free expansion is a classic example of a process that is *not* quasi-static, so $\int p \, dV$ cannot be used for it.

6. Thermal equilibrium, the Zeroth law and temperature scales

6.1 Thermal equilibrium and the Zeroth law

When a hot body and a cold body are brought into contact through a **diathermic** (heat-conducting) wall, changes occur in both until a state is reached in which no further change takes place; the bodies are then in **thermal equilibrium**. Bodies separated by an **adiabatic** wall (perfect insulator) do not interact thermally.

Zeroth law of thermodynamics. If two bodies are each in thermal equilibrium with a third body, they are in thermal equilibrium with each other.

The law is called the "zeroth" law because, although it was formulated after the first and second laws (by R. H. Fowler, 1931), it is logically prior to them. It is the basis of temperature measurement: the third body is the thermometer. If a thermometer reads the same value when placed in contact with bodies A and B, then A and B are in thermal equilibrium with each other and are said to have the same **temperature**. Temperature is thus the property that decides whether or not two bodies are in thermal equilibrium – heat flows from the body at the higher temperature to the body at the lower temperature until the temperatures are equal.

6.2 Thermometers and thermometric properties

A thermometer uses some measurable **thermometric property** that varies with temperature: the length of a mercury column (liquid-in-glass), the pressure of a gas at constant volume (constant-volume gas thermometer), the electrical resistance of a platinum wire (resistance thermometer), the e.m.f. of a thermocouple, or the intensity of radiation (pyrometer). The gas thermometer is the most fundamental, because all gases at low pressure give the same reading and it leads directly to the absolute (Kelvin) scale.

6.3 Temperature scales and the triple point

Modern temperature scales use a **single fixed point**: the triple point of water, the state at which ice, liquid water and water vapour coexist in equilibrium (0.01 °C, 0.6113 kPa). It is assigned the value **273.16 K** exactly on the Kelvin scale. The Kelvin (absolute, thermodynamic) scale is defined so that $T = 273.16 \times (p/p_{tp})$ for a constant-volume ideal-gas thermometer, where p_{tp} is the gas pressure at the triple point; its zero is the absolute zero of temperature. The commonly used scales are related as follows:

Scale	Absolute zero	Ice point	Triple point of water	Steam point
Kelvin (K)	0	273.15	273.16	373.15
Celsius (°C)	-273.15	0	0.01	100
Fahrenheit (°F)	-459.67	32	32.02	212
Rankine (°R)	0	491.67	491.69	671.67

Figure 1.2 Fixed points on the four common temperature scales.

$$T \text{ (K)} = t \text{ (°C)} + 273.15 \quad T \text{ (°R)} = t \text{ (°F)} + 459.67 \quad t \text{ (°F)} = 1.8 t \text{ (°C)} + 32 \quad T \text{ (°R)} = 1.8 T \text{ (K)}$$

A temperature *difference* of 1 K equals a difference of 1 °C, and a difference of 1 °R equals 1 °F. Every thermodynamic relation ($pV = mRT$, the Carnot efficiency, the entropy definition) requires **absolute** temperature; Celsius values must always be converted to kelvin before substitution.

7. Energy interactions – work and heat

Energy can cross the boundary of a closed system in only two forms: **work** and **heat**. Both are transient, boundary phenomena – they are not stored in the system – and both are path functions.

7.1 Work – the thermodynamic definition

In mechanics, work is done when a force moves its point of application: $W = \int F dx$. Thermodynamics uses a broader definition: *work is said to be done by a system on its surroundings if the sole effect external to the system could be reduced to the raising of a weight*. The weight need not actually be raised; it is enough that the effect *could* be so reduced. For example, when a battery drives a motor, the electrical energy crossing the boundary is work, because the motor could be used to wind up a weight. Conversely, if a battery heats a resistor immersed in water, the interaction between the battery and the resistor is work (the current could have driven a motor), but the interaction between the resistor and the water is heat.

7.2 Sign convention

Heat added to the system is positive; heat rejected by the system is negative.
Work done by the system is positive; work done on the system is negative.

With this convention the first law for a process reads $Q - W = \Delta E$. The convention is a historical one: the engine designer's interest was in heat supplied and work delivered.

7.3 Forms of work

- **Displacement ($p \, dV$) work** – the work of a moving boundary, treated in Section 8; this is the most important form for closed systems.
- **Paddle-wheel (stirring) work** – a paddle rotated inside a fluid by a falling weight does work on the fluid; the fluid volume is unchanged, so this work is *not* $p \, dV$ work and cannot be evaluated from the p – V diagram.
- **Shaft work** – the work transmitted by a rotating shaft, $W = 2\pi NT$ (T torque, N revolutions).
- **Electrical work** – $W = \int V I \, dt$ for a current I flowing across a potential difference V .
- **Flow work** – the work needed to push a fluid across the boundary of a control volume, $p v$ per kg (Section 12); it appears only in open systems.

7.4 Heat

Heat is energy in transit across the boundary of a system by virtue of a **temperature difference** between the system and its surroundings; it flows from the higher temperature to the lower. Heat transfer is a boundary phenomenon: a system does not contain heat, it contains energy. A process with no heat transfer (a perfectly insulated boundary, or a very rapid process) is **adiabatic**. Heat is measured in joules; the rate of heat transfer $\dot{Q}$ is in watts.

Heat is not temperature. Temperature is an intensive property of a body that measures the degree of hotness and decides the direction of heat flow; heat is the energy transferred because of a temperature difference. Two bodies at the same temperature exchange no heat, however large their energy content; a small body at a high temperature may contain far less energy than a large body at a lower temperature.

7.5 Comparison of heat and work

Aspect	Heat	Work
Nature	Energy transfer due to a temperature difference	Energy transfer by a force through a displacement (or its equivalent)
Molecular picture	Disorganised, random molecular energy transfer	Organised, macroscopic (directed) transfer
Recognised at the boundary?	Yes – exists only while crossing the boundary	Yes – exists only while crossing the boundary
Point or path function?	Path function (δQ inexact)	Path function (δW inexact)
Convertibility	Cannot be completely converted into work in a cycle (second law)	Can be completely converted into heat (friction, resistance heating)
Grade of energy	Low grade	High grade
Sign convention	+ into the system	+ out of the system

Both are transient phenomena, both are boundary phenomena, both have the same units and both are path functions; the essential difference is that work is an *organised* transfer that could raise a weight, whereas heat is a *disorganised* transfer driven only by a temperature difference.

8. Displacement work and the p – V diagram

8.1 Work of a moving boundary

Consider a gas at pressure p in a cylinder closed by a frictionless piston of area A . In a quasi-static process the gas pressure acts uniformly on the piston; when the piston moves an infinitesimal distance dx the force pA does work $\delta W = pA \, dx = p \, dV$. For a finite quasi-static process from state 1 to state 2,

$$W_{1-2} = \int_1^2 p \, dV = \text{area under the process curve on the } p\text{--}V \text{ diagram}$$

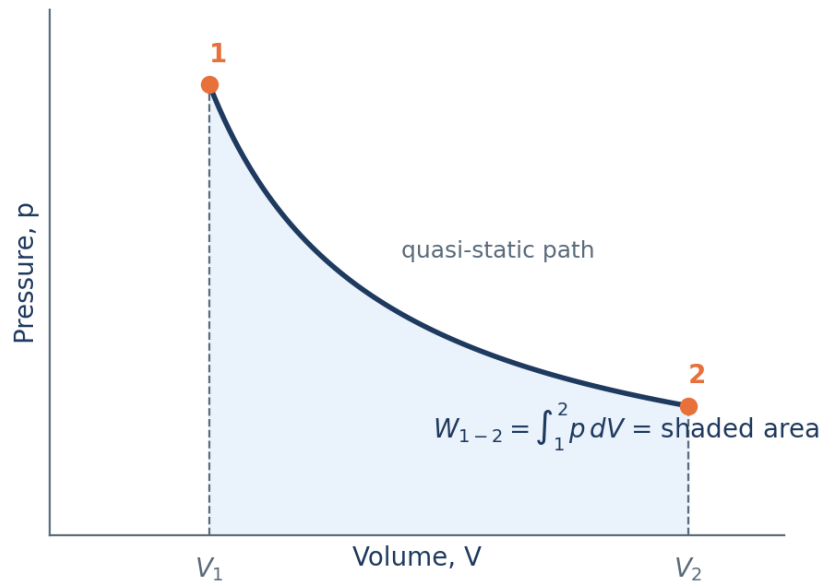


Figure 1.3 Displacement work: the area under the quasi-static path on the p – V diagram.

Three conditions must hold for this expression: the process must be quasi-static (so that p is defined and uniform), the boundary must move ($dV \neq 0$), and p must be the pressure of the *system*. It gives positive work for expansion ($dV > 0$, work done by the gas) and negative work for compression.

8.2 Work is a path function

The area under the curve depends on the curve, not merely on its end points. Figure 1.4 shows two quasi-static paths A and B between the same states 1 and 2; the areas – and therefore the work transfers – differ. This is the graphical proof that work is a path function: δW is an inexact differential and $\oint \delta W \neq 0$ for a cycle (the enclosed area is the net work of the cycle).

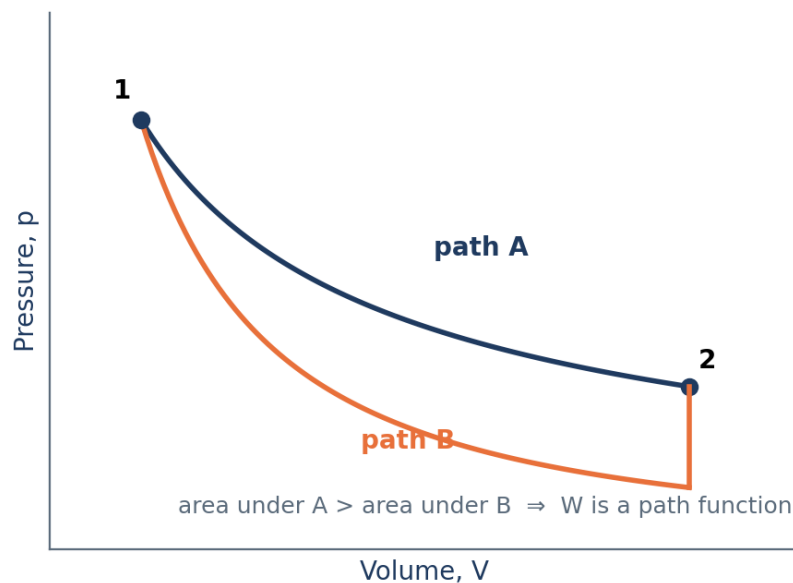


Figure 1.4 Two paths between the same end states enclose different areas – work is a path function.

8.3 Work in the standard quasi-static processes

For a fixed mass of ideal gas ($pV = mRT$) the common processes and their work are derived below; Figure 1.5 shows them on the p – V plane, all starting from the same state 1. In the polytropic law $pV^n = C$ the

index n takes the value 0 for constant pressure, 1 for isothermal, $\gamma = c_p/c_v$ for reversible adiabatic, ∞ for constant volume, and typically 1.2–1.35 for real compressions and expansions.

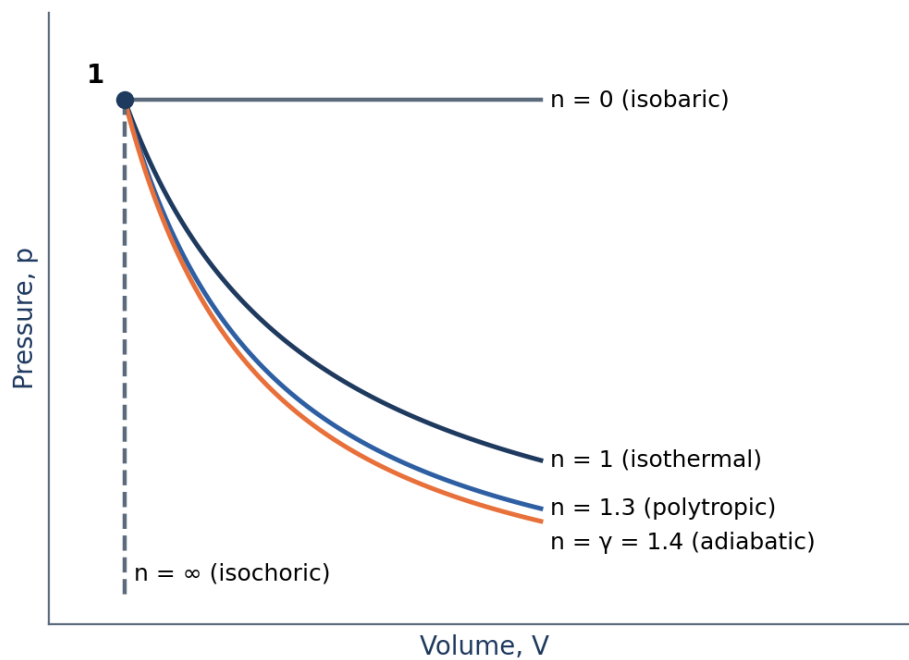


Figure 1.5 The family of polytropic processes $pV^n = C$ on the p – V diagram.

(a) **Constant pressure (isobaric), $n = 0$.** $W_{1-2} = p \int dV = p(V_2 - V_1) = mR(T_2 - T_1)$.

(b) **Constant volume (isochoric), $n = \infty$.** $dV = 0$, so $W_{1-2} = 0$. (Paddle or electrical work may still be done on the gas, but there is no displacement work.)

(c) **Isothermal, $pV = C$, $n = 1$.** $p = C/V$, so $W_{1-2} = C \int dV/V = p_1 V_1 \ln(V_2/V_1) = mRT \ln(V_2/V_1) = p_1 V_1 \ln(p_1/p_2)$.

(d) **Polytropic, $pV^n = C$ ($n \neq 1$).** $p = C V^{-n}$, so $W_{1-2} = C \int V^{-n} dV = C [V^{1-n}]/(1 - n)$ between V_1 and V_2 . Since $C = p_1 V_1^n = p_2 V_2^n$,

$$W_{1-2} = (p_1 V_1 - p_2 V_2)/(n - 1) = mR(T_1 - T_2)/(n - 1)$$

Useful relations for the polytropic process: $T_2/T_1 = (V_1/V_2)^{n-1} = (p_2/p_1)^{(n-1)/n}$, and the index from two states, $n = \ln(p_2/p_1)/\ln(V_1/V_2)$.

(e) **Reversible adiabatic, $pV^\gamma = C$.** Same as (d) with $n = \gamma$: $W_{1-2} = (p_1 V_1 - p_2 V_2)/(\gamma - 1) = mR(T_1 - T_2)/(\gamma - 1) = mc_v(T_1 - T_2)$. (This also follows from the first law with $Q = 0$: $W = -\Delta U$.)

KEY POINT The p – V diagram shows displacement work only. Work done on a fluid by a paddle, a shaft or an electric heater does not appear as an area, and the p – V area is meaningful only for quasi-static processes. In a free expansion the area under the (undefined) path is *not* the work – the work is zero.

9. The first law of thermodynamics for closed systems

9.1 Joule's experiments and the first law for a cycle

In his paddle-wheel experiments (1840–49) Joule stirred water in an insulated vessel with a paddle driven by falling weights, measured the work done and the rise in temperature, and then allowed the water to cool back to its initial temperature by heat transfer to the surroundings. Over the complete cycle he found

that the work supplied was always proportional to the heat rejected, whatever the amount or the manner of the work – the famous mechanical equivalent of heat, $4.186 \text{ kJ} = 1 \text{ kcal}$. This is the first law for a closed system undergoing a **cycle**:

$$\oint \delta Q = \oint \delta W \quad (\text{the net heat transfer equals the net work transfer, in consistent units})$$

9.2 The first law for a process – energy is a property

Consider a closed system that goes from state 1 to state 2 by path A and returns by path B (Figure 1.6). For the cycle 1-A-2-B-1, $\oint \delta Q - \oint \delta W = 0$, i.e. $(Q_A - W_A) + (Q_B - W_B) = 0$. Now let the system return instead by a different path C: $(Q_A - W_A) + (Q_C - W_C) = 0$. Subtracting, $(Q_B - W_B) = (Q_C - W_C)$: the quantity $Q - W$ has the same value for every path between the same two states. It therefore depends only on the end states and is the change of a **property**, which we call the **energy** E of the system:

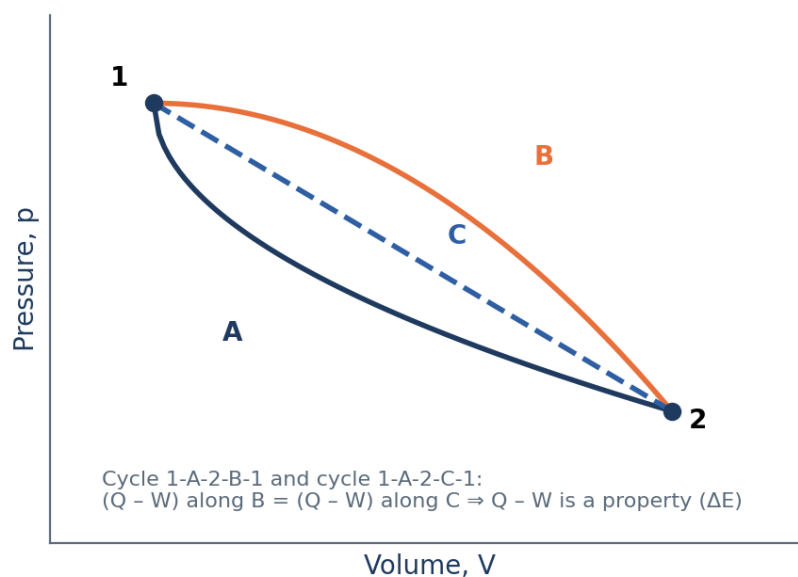


Figure 1.6 Two return paths B and C give the same value of $Q - W$, so $Q - W$ is the change of a property.

$$\text{First law for a process: } Q - W = \Delta E = E_2 - E_1, \quad \text{where } E = U + KE + PE$$

For a stationary system (no change in kinetic or potential energy) this becomes $Q - W = \Delta U$, or in differential form $\delta Q - \delta W = dU$. The equation says that energy is conserved: the heat added to a system minus the work done by it is stored as an increase in its energy.

Corollaries. (i) *Internal energy is a property of the system (shown above).* (ii) *The energy of an isolated system remains constant:* with $Q = 0$ and $W = 0$, $\Delta E = 0$ – energy can be neither created nor destroyed, only converted from one form to another. (iii) *A perpetual motion machine of the first kind (PMM1) is impossible.* A PMM1 is a device that would deliver work continuously without any energy supply – it would create energy, violating the first law. The converse is also true: a machine that continuously absorbs work without any other effect is equally impossible.

9.3 Internal energy

The **internal energy** U is the energy stored in a system by virtue of the motion and configuration of its molecules – the translational, rotational and vibrational kinetic energy of the molecules, and the potential energy of the intermolecular forces – as distinct from the macroscopic kinetic and potential energy of the system as a whole. It is an extensive property and a point function; its change $\Delta U = Q - W$ depends only

on the end states, and $\oint dU = 0$. For an ideal gas U depends on temperature alone (Joule's law). The specific internal energy $u = U/m$ has the unit kJ/kg.

9.4 Limitations of the first law

The first law is a statement of energy conservation and nothing more. It has important limitations, which are removed by the second law (Unit II):

- It gives no information about the **direction** in which a process can occur: it would allow heat to flow from a cold body to a hot body, or a stone to cool and rise into the air, provided energy were conserved.
- It makes no distinction between heat and work as forms of energy, treating them as fully interchangeable; in reality work can be completely converted to heat but heat cannot be completely converted to work in a cycle.
- It does not indicate whether a process is possible at all, nor the extent to which a process will proceed before equilibrium is reached, nor the conditions of equilibrium.
- It does not tell us how much of the energy in a system is *available* for producing work – it says nothing about the *quality* of energy.

10. Internal energy, enthalpy and specific heats

10.1 Enthalpy

In many processes – constant-pressure heating in a closed system, and all flow processes – the combination $U + pV$ occurs naturally. It is given the name **enthalpy**:

$$H = U + pV \quad (\text{kJ}) \quad h = u + pv \quad (\text{kJ/kg})$$

Enthalpy is a property because U , p and V are properties. For a constant-pressure quasi-static process in a closed system, $Q = \Delta U + p(V_2 - V_1) = (U_2 + pV_2) - (U_1 + pV_1) = \Delta H$: the heat added at constant pressure equals the increase in enthalpy. For an ideal gas $h = u + RT$, so h also depends on temperature alone.

10.2 Specific heats

The **specific heat at constant volume** is the heat required to raise the temperature of unit mass by one degree while the volume is held constant. Since no displacement work is done at constant volume, $\delta Q = dU$ and

$$c_v = (\partial u / \partial T)_v \quad c_p = (\partial h / \partial T)_p \quad (\text{kJ/kg K})$$

The **specific heat at constant pressure** is the heat required to raise the temperature of unit mass by one degree at constant pressure; because $\delta Q = dH$ at constant pressure, c_p is the derivative of h with respect to T . For an ideal gas u and h depend on T only, so the partial derivatives become ordinary ones: $du = c_v dT$ and $dh = c_p dT$ for *any* process, not only at constant volume or constant pressure. With constant specific heats, $\Delta U = mc_v(T_2 - T_1)$ and $\Delta H = mc_p(T_2 - T_1)$.

10.3 Relation between c_p , c_v and R for an ideal gas

From $h = u + pv = u + RT$, differentiating with respect to T : $dh/dT = du/dT + R$, i.e.

$$c_p - c_v = R \quad \text{and} \quad \gamma = c_p/c_v, \quad \text{so} \quad c_v = R/(\gamma - 1), \quad c_p = \gamma R/(\gamma - 1)$$

c_p is greater than c_v because at constant pressure part of the heat supplied goes into the expansion work $p dV$, whereas at constant volume all of it raises the internal energy. For air, $R = 0.287$ kJ/kg K, $c_p = 1.005$

kJ/kg K, $c_v = 0.718$ kJ/kg K and $\gamma = 1.4$; monatomic gases have $\gamma = 1.67$ and most diatomic gases about 1.4.

11. Application of the first law to closed-system processes

Combining the first law ($Q = \Delta U + W$) with the ideal-gas relations and the work expressions of Section 8 gives the complete energy accounting for each standard non-flow process. Table 1.1 summarises the results for a mass m of ideal gas with constant specific heats.

Process	Law	Work W_{1-2}	$\Delta U = mc_v(T_2 - T_1)$	Heat Q_{1-2}
Constant volume	$V = C$	0	$mc_v(T_2 - T_1)$	$mc_v(T_2 - T_1) = \Delta U$
Constant pressure	$p = C$	$p(V_2 - V_1) = mR(T_2 - T_1)$	$mc_v(T_2 - T_1)$	$mc_p(T_2 - T_1) = \Delta H$
Isothermal	$pV = C$	$p_1V_1 \ln(V_2/V_1)$	0	$= W$ (all heat goes to work)
Polytropic	$pV^n = C$	$(p_1V_1 - p_2V_2)/(n - 1)$	$mc_v(T_2 - T_1)$	$\Delta U + W = W(\gamma - n)/(\gamma - 1)$
Reversible adiabatic	$pV^\gamma = C$	$(p_1V_1 - p_2V_2)/(\gamma - 1) = -\Delta U$	$mc_v(T_2 - T_1)$	0

The polytropic heat transfer, $Q = W(\gamma - n)/(\gamma - 1)$, is worth remembering: for $1 < n < \gamma$ (the usual range for compression with cooling) the heat is negative during compression – part of the work is rejected as heat – and the process lies *between* the isothermal and the adiabatic curves on the p – V diagram.

11.1 Multi-process cycles – the first-law table

When a closed system executes a cycle of several processes, the first law applied to each process ($Q = \Delta E + W$) and to the whole cycle ($\sum Q = \sum W$, $\sum \Delta E = 0$) allows every unknown in a table of Q , W and ΔE to be found. The method is: (1) for each process with two known quantities, find the third from $Q = \Delta E + W$; (2) use $\sum \Delta E = 0$ over the cycle to find the missing ΔE ; (3) use $\sum Q = \sum W$ (or the given cyclic total) to find the missing Q or W . Worked Example 1.1 shows the procedure.

WORKED EXAMPLE 1.1 — First-law table for a four-process cycle

Problem. A piston–cylinder machine contains a fluid system that passes through a cycle of four processes, a–b, b–c, c–d and d–a. During each cycle the sum of all heat transfers is –180 kJ, and the machine completes 120 cycles per minute. Complete the following table (kJ/min), stating the method used for each entry, and determine the net rate of work output in kW.

Process	Q (kJ/min)	W (kJ/min)	ΔE (kJ/min)
a – b	0	2400	—
b – c	22000	0	—
c – d	–2200	—	–35000
d – a	—	—	—

Given: $\sum Q = -180$ kJ per cycle; 120 cycles/min; table entries as above.

Formula: First law for each process, $Q = \Delta E + W$; for the cycle, $\sum \Delta E = 0$ and $\sum Q = \sum W$.

Solution.

Cyclic heat transfer per minute: $\sum Q = -180 \times 120 = -21600$ kJ/min.

Process a–b: $\Delta E = Q - W = 0 - 2400 = -2400$ kJ/min.

Process b–c: $\Delta E = Q - W = 22000 - 0 = 22000$ kJ/min.

Process c–d: $W = Q - \Delta E = -2200 - (-35000) = 32800$ kJ/min.

Process d–a: $Q_{da} = \Sigma Q - (Q_{ab} + Q_{bc} + Q_{cd}) = -21600 - (0 + 22000 - 2200) = -41400 \text{ kJ/min.}$

$\Delta E_{da} = -(\Delta E_{ab} + \Delta E_{bc} + \Delta E_{cd}) = -(-2400 + 22000 - 35000) = 15400 \text{ kJ/min.}$

$W_{da} = Q_{da} - \Delta E_{da} = -41400 - 15400 = -56800 \text{ kJ/min.}$

Process	Q (kJ/min)	W (kJ/min)	ΔE (kJ/min)	Method
a – b	0	2400	-2400	$\Delta E = Q - W$
b – c	22000	0	22000	$\Delta E = Q - W$
c – d	-2200	32800	-35000	$W = Q - \Delta E$
d – a	-41400	-56800	15400	$\Sigma Q, \Sigma \Delta E = 0, W = Q - \Delta E$

Check: $\Sigma W = 2400 + 0 + 32800 - 56800 = -21600 \text{ kJ/min} = \Sigma Q \checkmark.$

Net rate of work = $-21600 \text{ kJ/min} = -21600/60 \text{ kW}.$

ANSWER Net work output = -360 kW , i.e. the machine *absorbs* 360 kW (it is a work-consuming device such as a compressor or refrigerator).

11.2 Cycles made up of standard processes

WORKED EXAMPLE 1.2 — Isothermal – constant volume – polytropic cycle

Problem. An ideal gas occupying 0.2 m^3 at 12 bar and 500 K is expanded isothermally to four times its initial volume. It is then cooled at constant volume to 300 K and finally compressed polytropically back to the initial state. Sketch the cycle on the p – V plane and calculate (i) the index of the polytropic compression, (ii) the net work done and (iii) the net heat transfer for the cycle.

Given: $p_1 = 12 \text{ bar} = 1200 \text{ kPa}$, $V_1 = 0.2 \text{ m}^3$, $T_1 = 500 \text{ K}$; $V_2 = 4V_1$; $T_3 = 300 \text{ K}$.

Formula: Isothermal: $p_1V_1 = p_2V_2$, $W = p_1V_1 \ln(V_2/V_1)$. Constant volume: $p_3/p_2 = T_3/T_2$, $W = 0$.

Polytropic: $p_3V_3^n = p_1V_1^n$, $W = (p_3V_3 - p_1V_1)/(n - 1)$. Cycle: $\Sigma Q = \Sigma W$.

Solution.

Process 1–2 (isothermal): $V_2 = 0.8 \text{ m}^3$; $p_2 = p_1V_1/V_2 = 12/4 = 3 \text{ bar}$.

$W_{1-2} = 1200 \times 0.2 \times \ln 4 = 240 \times 1.3863 = 332.7 \text{ kJ}.$

Process 2–3 (constant volume): $p_3 = p_2 \times T_3/T_2 = 3 \times 300/500 = 1.8 \text{ bar}$; $W_{2-3} = 0$.

Process 3–1 (polytropic): $(V_3/V_1)^n = p_1/p_3 \Rightarrow 4^n = 12/1.8 = 6.667 \Rightarrow n = \ln 6.667 / \ln 4 = 1.897/1.386 = 1.368$.

$W_{3-1} = (p_3V_3 - p_1V_1)/(n - 1) = (180 \times 0.8 - 1200 \times 0.2)/0.368 = (144 - 240)/0.368 = -260.5 \text{ kJ}.$

Net work: $W_{\text{net}} = 332.7 + 0 - 260.5 = 72.2 \text{ kJ}.$

Net heat: $Q_{\text{net}} = W_{\text{net}} = 72.2 \text{ kJ}$ (first law for a cycle).

ANSWER (i) $n = 1.37$; (ii) $W_{\text{net}} = 72.2 \text{ kJ}$ done by the gas; (iii) $Q_{\text{net}} = 72.2 \text{ kJ}$ supplied to the gas.

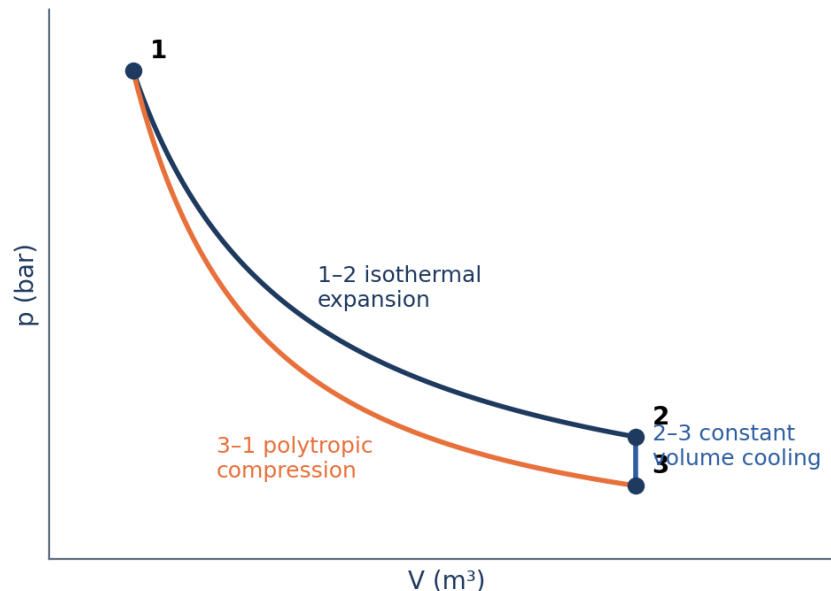


Figure 1.7 The cycle of Worked Example 1.2 on the p - V plane.

11.3 Compression of a gas – index, internal energy and heat

WORKED EXAMPLE 1.3 — Polytropic compression in a cylinder

Problem. A cylinder contains 0.4 m^3 of a gas at 100 kPa and 75°C . The gas is compressed to 0.12 m^3 , the final pressure being 500 kPa . Take $\gamma = 1.4$ and $R = 0.29 \text{ kJ/kg K}$. Determine (i) the mass of gas, (ii) the index n of the compression, (iii) the change in internal energy, (iv) the heat transferred, stating whether it is received or rejected, and (v) whether the process lies between the isothermal and the reversible adiabatic compression of the same gas.

Given: $V_1 = 0.4 \text{ m}^3$, $p_1 = 100 \text{ kPa}$, $T_1 = 348.15 \text{ K}$; $V_2 = 0.12 \text{ m}^3$, $p_2 = 500 \text{ kPa}$; $\gamma = 1.4$, $R = 0.29 \text{ kJ/kg K} \Rightarrow c_v = R/(\gamma - 1) = 0.725 \text{ kJ/kg K}$.

Formula: $m = p_1 V_1 / (RT_1)$; $n = \ln(p_2/p_1) / \ln(V_1/V_2)$; $T_2 = p_2 V_2 / (mR)$; $\Delta U = mc_v(T_2 - T_1)$; $W = (p_1 V_1 - p_2 V_2) / (n - 1)$; $Q = \Delta U + W$.

Solution.

(i) $m = 100 \times 0.4 / (0.29 \times 348.15) = 40/100.96 = 0.396 \text{ kg}$.

(ii) $n = \ln(500/100) / \ln(0.4/0.12) = 1.609/1.204 = 1.337$.

Final temperature: $T_2 = 500 \times 0.12 / (0.396 \times 0.29) = 60/0.1149 = 522.2 \text{ K}$.

(iii) $\Delta U = 0.396 \times 0.725 \times (522.2 - 348.15) = 0.2872 \times 174.1 = +50.0 \text{ kJ}$.

Work: $W = (100 \times 0.4 - 500 \times 0.12) / (1.337 - 1) = (40 - 60) / 0.337 = -59.4 \text{ kJ}$ (work done on the gas = 59.4 kJ).

(iv) $Q = \Delta U + W = 50.0 - 59.4 = -9.4 \text{ kJ} \rightarrow$ heat is rejected.

(v) Isothermal compression has $n = 1$, reversible adiabatic has $n = \gamma = 1.4$; here $n = 1.337$ lies between them, so the process lies between the isothermal and adiabatic curves on the p - V plane: part of the compression work (9.4 of 59.4 kJ) is rejected as heat and the rest raises the internal energy – consistent with $Q = W(\gamma - n) / (\gamma - 1) = -59.4 \times 0.063 / 0.4 = -9.4 \text{ kJ} \checkmark$.

ANSWER $m = 0.396 \text{ kg}$; $n = 1.34$; $\Delta U = +50.0 \text{ kJ}$; $Q = -9.4 \text{ kJ}$ (rejected); the process lies between isothermal and adiabatic.

11.4 Two processes in series and the equivalent polytropic process

WORKED EXAMPLE 1.4 — Adiabatic expansion followed by constant-pressure heating

Problem. A closed system contains 0.25 m^3 of air at 5 bar and 140°C . The air expands reversibly and adiabatically until the pressure falls to 1.05 bar, and is then heated at constant pressure until its enthalpy has increased by 80 kJ. Calculate the total work done, and find the index of expansion of a single reversible polytropic process between the same initial and final states that would give the same work. Take $c_p = 1.005 \text{ kJ/kg K}$ and $c_v = 0.718 \text{ kJ/kg K}$.

Given: $V_1 = 0.25 \text{ m}^3$, $p_1 = 500 \text{ kPa}$, $T_1 = 413.15 \text{ K}$; $p_2 = 105 \text{ kPa}$; $\Delta H_{2-3} = 80 \text{ kJ}$; $c_p = 1.005$, $c_v = 0.718 \Rightarrow R = 0.287 \text{ kJ/kg K}$, $\gamma = 1.4$.

Formula: $m = p_1 V_1 / (RT_1)$; $T_2 = T_1 (p_2/p_1)^{(\gamma-1)/\gamma}$; $W_{1-2} = mR(T_1 - T_2)/(\gamma - 1)$; $\Delta H = mc_p(T_3 - T_2)$; $W_{2-3} = p(V_3 - V_2) = mR(T_3 - T_2)$; polytropic 1–3: $W = mR(T_1 - T_3)/(n - 1)$.

Solution.

$m = 500 \times 0.25 / (0.287 \times 413.15) = 125/118.6 = 1.054 \text{ kg}$.

Process 1–2: $T_2 = 413.15 \times (105/500)^{0.2857} = 413.15 \times 0.6403 = 264.6 \text{ K}$.

$W_{1-2} = 1.054 \times 0.287 \times (413.15 - 264.6)/0.4 = 112.5 \text{ kJ}$.

Process 2–3: $T_3 - T_2 = 80 / (1.054 \times 1.005) = 75.5 \text{ K} \Rightarrow T_3 = 340.1 \text{ K}$.

$W_{2-3} = 1.054 \times 0.287 \times 75.5 = 22.8 \text{ kJ}$.

Total work = $112.5 + 22.8 = 135.3 \text{ kJ}$.

Equivalent polytropic 1–3: $135.3 = 1.054 \times 0.287 \times (413.15 - 340.1)/(n - 1) = 22.1/(n - 1) \Rightarrow n - 1 = 0.163$.

ANSWER Total work = 135.3 kJ; equivalent polytropic index $n \approx 1.16$ (close to 1 because the overall change 1 $\rightarrow$ 3 is nearly isothermal – a large amount of work is done for a small net temperature change).

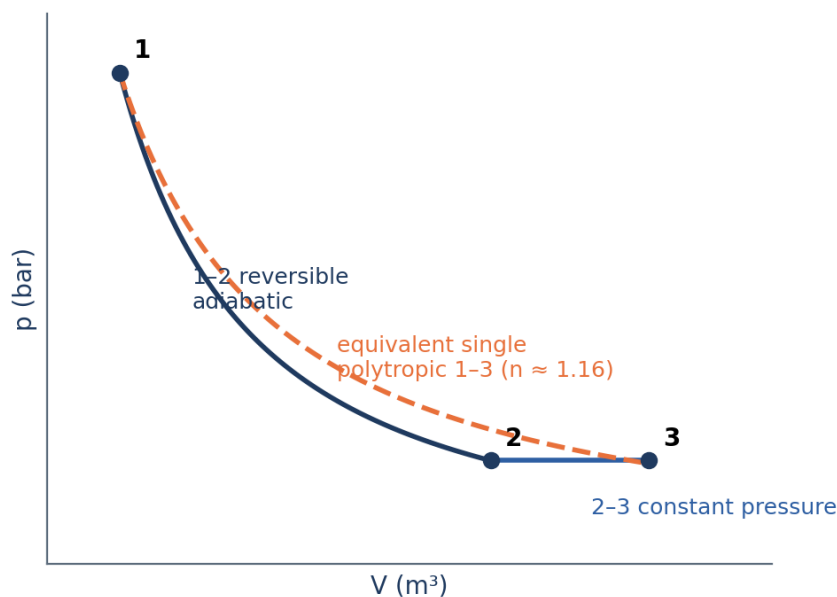


Figure 1.8 Worked Example 1.4: adiabatic expansion 1–2, constant-pressure heating 2–3, and the equivalent polytropic 1–3.

11.5 Verifying the first law for a cycle

WORKED EXAMPLE 1.5 — Three-process cycle with $\Sigma Q = \Sigma W$ check

Problem. A gas undergoes a thermodynamic cycle consisting of: process 1–2, constant pressure $p = 1.5 \text{ bar}$, $V_1 = 0.03 \text{ m}^3$, $W_{1-2} = 12 \text{ kJ}$; process 2–3, compression with $pV = \text{constant}$, $U_3 = U_2$; process 3–1, constant volume, $U_1 - U_3 = -28 \text{ kJ}$. Changes in kinetic and potential energy are

negligible. (i) Sketch the cycle on the p–V plane; (ii) calculate the net work of the cycle; (iii) calculate the heat transfer for process 1–2; (iv) show that $\Sigma Q = \Sigma W$ for the cycle.

Given: $p_1 = p_2 = 150 \text{ kPa}$, $V_1 = 0.03 \text{ m}^3$, $W_{1-2} = 12 \text{ kJ}$; $V_3 = V_1$; $U_3 = U_2$; $U_1 - U_3 = -28 \text{ kJ}$.

Formula: $W_{1-2} = p(V_2 - V_1)$; $W_{2-3} = p_2 V_2 \ln(V_3/V_2)$; $W_{3-1} = 0$; $Q = \Delta U + W$ for each process.

Solution.

(i) The cycle is a horizontal line 1–2 (expansion), a rectangular hyperbola 2–3 rising to the left, and a vertical line 3–1 down to state 1 (Figure 1.9).

Process 1–2: $V_2 = V_1 + W_{1-2}/p = 0.03 + 12/150 = 0.11 \text{ m}^3$.

Process 2–3: $V_3 = 0.03 \text{ m}^3$, $p_3 = p_2 V_2/V_3 = 150 \times 0.11/0.03 = 550 \text{ kPa} = 5.5 \text{ bar}$.

$W_{2-3} = 150 \times 0.11 \times \ln(0.03/0.11) = 16.5 \times (-1.299) = -21.4 \text{ kJ}$.

Process 3–1: $W_{3-1} = 0$.

(ii) $W_{\text{net}} = 12 - 21.4 + 0 = -9.4 \text{ kJ}$ (net work done *on* the gas).

(iii) $U_2 - U_1 = -(U_1 - U_3) = +28 \text{ kJ}$ (since $U_3 = U_2$); $Q_{1-2} = \Delta U + W = 28 + 12 = 40 \text{ kJ}$.

(iv) $Q_{2-3} = (U_3 - U_2) + W_{2-3} = 0 - 21.4 = -21.4 \text{ kJ}$; $Q_{3-1} = (U_1 - U_3) + 0 = -28 \text{ kJ}$.

$\Sigma Q = 40 - 21.4 - 28 = -9.4 \text{ kJ} = \Sigma W \checkmark$.

ANSWER $W_{\text{net}} = -9.4 \text{ kJ}$; $Q_{1-2} = 40 \text{ kJ}$; $\Sigma Q = \Sigma W = -9.4 \text{ kJ}$, so the first law is satisfied.

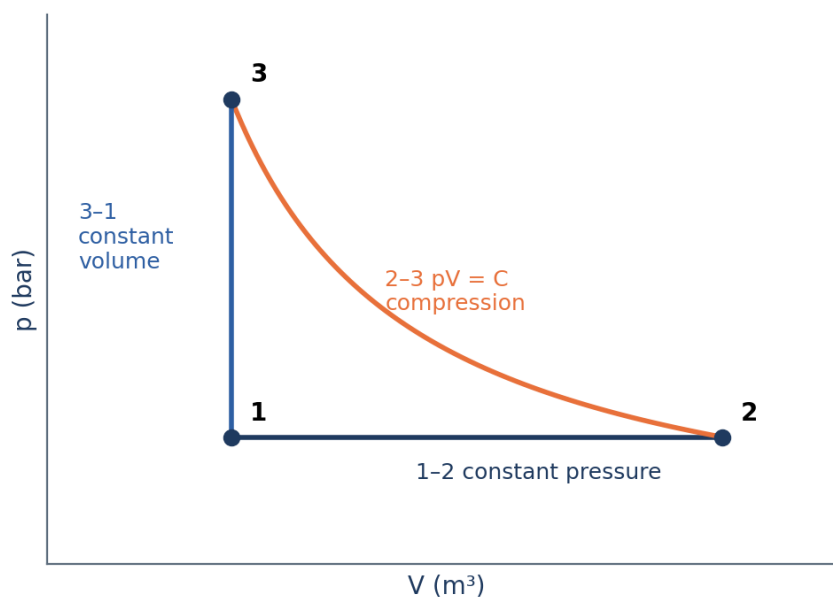


Figure 1.9 The three-process cycle of Worked Example 1.5.

12. The first law for open systems – the steady flow energy equation

Most engineering equipment – turbines, compressors, pumps, boilers, nozzles – involves a continuous flow of fluid, so the analysis must be made for an open system or **control volume** (CV). The control volume is a fixed region in space bounded by a **control surface**, through which mass enters and leaves carrying its energy with it. In addition to heat and shaft work crossing the control surface, a new form of work appears: the work required to push the fluid across the boundary.

12.1 Flow work

Consider the fluid about to enter the control volume at section 1 (Figure 1.10). To push an element of mass dm (occupying volume $dm \cdot v_1$) into the CV against the pressure p_1 , the fluid behind it must do work

equal to force $\times$ distance $= (p_1 A_1)(dx_1) = p_1 dm v_1$. This is the **flow work** (or flow energy): **$p v$ per kg of fluid** at the section. Similarly, the fluid leaving at section 2 does flow work $p_2 v_2$ on the fluid ahead of it. Flow work is not a property of the fluid – it is the work associated with pushing the fluid across a boundary – and it appears only in open systems.

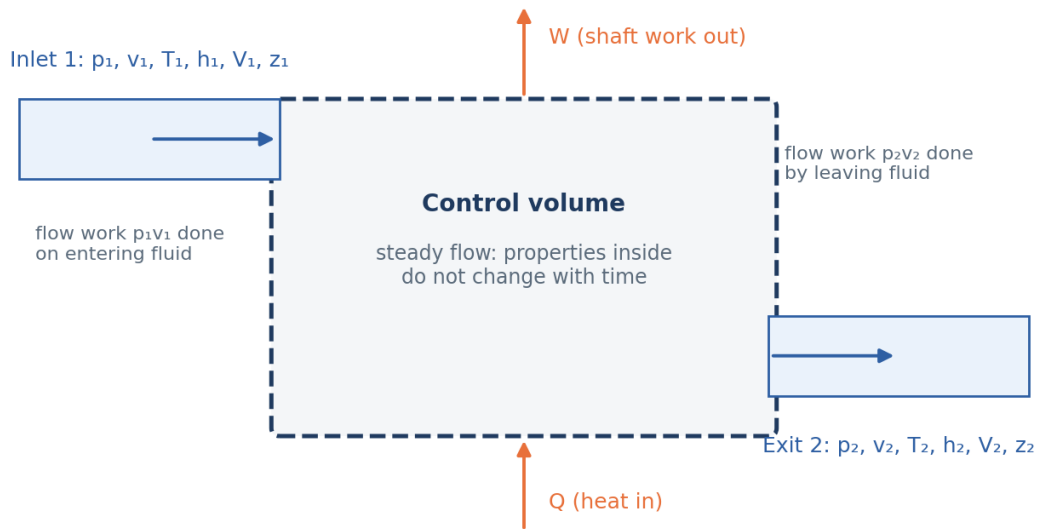


Figure 1.10 Control volume for the derivation of the steady flow energy equation.

12.2 Assumptions of steady flow

- The mass flow rate is constant and the same at inlet and exit: $\dot{m}_1 = \dot{m}_2 = \dot{m}$ (no accumulation of mass).
- The state of the fluid and its velocity at any point of the inlet and exit sections do not change with time (one-dimensional, uniform flow across each section).
- The rates of heat transfer $\dot{Q}$ and work transfer $\dot{W}$ are constant.
- No property within the control volume changes with time (no accumulation of energy).
- There is a single inlet and a single exit (single stream); the fluid crosses the control surface only at these sections.

12.3 Derivation of the SFEE

Apply the first law to the control volume over a small time interval dt during which mass dm enters at 1 and an equal mass dm leaves at 2. The energy carried by each kg of fluid consists of its internal energy u , its kinetic energy $V^2/2$ and its potential energy gz . The energy balance is:

energy entering with the fluid + flow work done on the entering fluid + heat added = energy leaving with the fluid + flow work done by the leaving fluid + shaft work delivered.

Per kg of fluid:

$$u_1 + p_1 v_1 + V_1^2/2 + gz_1 + q = u_2 + p_2 v_2 + V_2^2/2 + gz_2 + w.$$

The combination $u + pv$ occurs at each section; this is the enthalpy h . Hence the **steady flow energy equation (SFEE)**:

$$h_1 + V_1^2/2 + gz_1 + q = h_2 + V_2^2/2 + gz_2 + w \quad (\text{per kg; } h \text{ in J/kg, } V \text{ in m/s, } z \text{ in m})$$

Multiplying by $\dot{m}$ gives the time-rate form, $\dot{Q} + \dot{m}(h_1 + V_1^2/2 + gz_1) = \dot{W} + \dot{m}(h_2 + V_2^2/2 + gz_2)$. The continuity equation $\dot{m} = A_1 V_1/v_1 = A_2 V_2/v_2$ accompanies the SFEE and links the pipe areas, velocities and specific volumes.

Meaning of the terms. h – enthalpy, the stored internal energy plus the flow work needed to push the fluid across the boundary; $V^2/2$ – kinetic energy per kg; gz – potential energy per kg above a datum; q –

heat transfer to the fluid per kg flowing ($\dot{Q}/\dot{m}$); w – shaft work delivered by the fluid per kg flowing ($\dot{W}/\dot{m}$). The flow work has been absorbed into the enthalpy, which is why enthalpy, not internal energy, is the natural energy variable of flow processes.

Units. With h in kJ/kg the kinetic-energy term must be written $V^2/2000$ kJ/kg and the potential-energy term $gz/1000$ kJ/kg. A velocity change from 45 m/s to 50 m/s changes the kinetic energy by only $(50^2 - 45^2)/2000 = 0.24$ kJ/kg, and a 10 m rise changes the potential energy by 0.098 kJ/kg – negligible compared with enthalpy changes of hundreds of kJ/kg. Kinetic energy becomes important only at high velocities (nozzles, jets, exhausts above about 100 m/s), and potential energy only for large elevation differences (hydraulic machines, tall cooling towers).

KEY POINT Energy equation of a closed system versus the SFEE: for a closed system $Q - W = \Delta U$ (plus ΔKE and ΔPE if the system moves); for an open system in steady flow the internal energy is replaced by the enthalpy (because of the flow work pv), the changes are between inlet and exit rather than between initial and final states, and the kinetic and potential energy of the *flowing stream* appear explicitly.

13. Applications of the SFEE to engineering devices

The SFEE is applied to each device by identifying which terms are negligible (Figure 1.11). In every case below the potential-energy change is negligible.

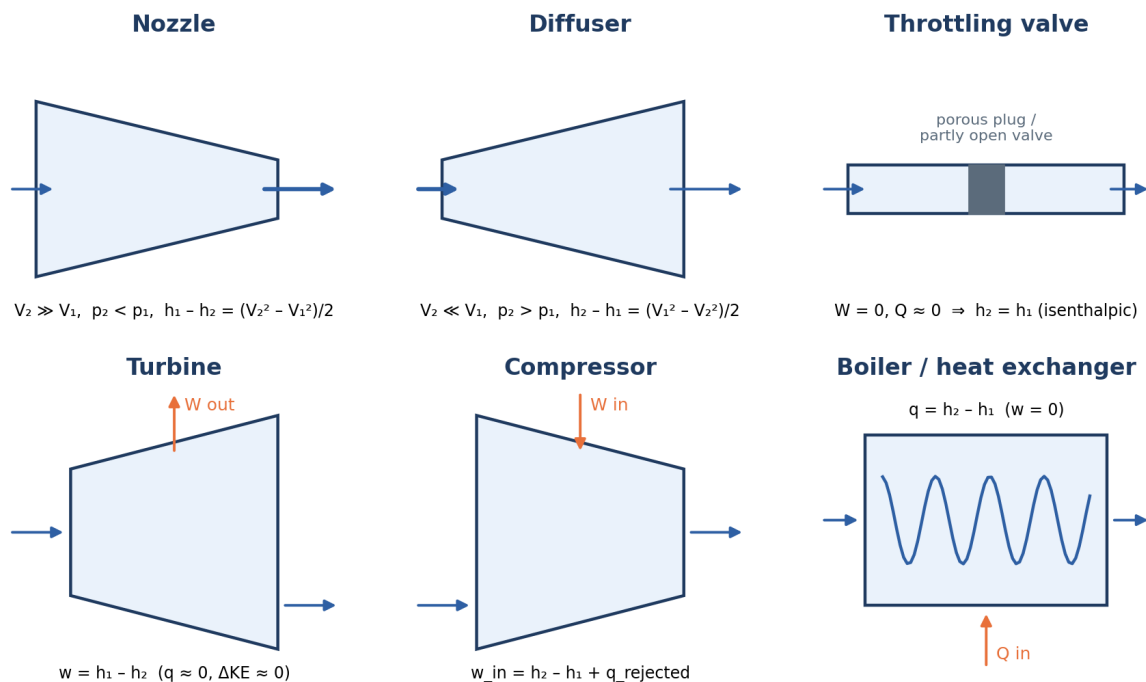


Figure 1.11 Steady-flow devices and the reduced form of the SFEE for each.

13.1 Nozzle and diffuser

A **nozzle** is a duct of decreasing cross-section that increases the velocity of a fluid at the expense of its pressure (and enthalpy); a **diffuser** is a duct of increasing cross-section that decelerates the fluid and raises its pressure. Both are short passages with no moving parts: $w = 0$, and the heat transfer is negligible because the fluid spends very little time inside ($q \approx 0$). The SFEE reduces to

$$h_1 + V_1^2/2 = h_2 + V_2^2/2 \quad \Rightarrow \quad \text{nozzle exit velocity } V_2 = \sqrt{2(h_1 - h_2) + V_1^2}$$

For the nozzle the enthalpy drop is converted into kinetic energy; if the inlet velocity is negligible, $V_2 = \sqrt{2(h_1 - h_2)}$ with h in J/kg. For the diffuser $h_2 - h_1 = (V_1^2 - V_2^2)/2$: the kinetic energy is converted into enthalpy and, for a gas, into a pressure rise ($T_2 = T_1 + (V_1^2 - V_2^2)/(2c_p)$, with p_2 then following from the isentropic relation). The exit area follows from continuity, $A_2 = \dot{m}v_2/V_2$.

13.2 Turbine and compressor

A **turbine** produces shaft work from the expansion of a fluid; the casing is insulated ($q \approx 0$), and the inlet and exit velocities are usually small or nearly equal ($\Delta KE \approx 0$), so $w = h_1 - h_2$ and the power is $\dot{W} = \dot{m}(h_1 - h_2)$. If the velocities are appreciable, $w = (h_1 - h_2) + (V_1^2 - V_2^2)/2$; if the casing loses heat, $w = (h_1 - h_2) - q_{\text{loss}}$.

A **compressor** (or pump) receives shaft work and raises the pressure of the fluid; w is negative. For a rotary compressor $q \approx 0$ and $\Delta KE \approx 0$, so the **work input** $= h_2 - h_1 = c_p(T_2 - T_1)$ for a gas. For a water-jacketed reciprocating compressor the heat rejected to the cooling water must be included: work input $= (h_2 - h_1) + q_{\text{rejected}}$, where $h_2 - h_1 = (u_2 - u_1) + (p_2v_2 - p_1v_1)$.

13.3 Boiler, condenser and heat exchanger

There is no shaft work, and the velocities and elevation changes are small, so $q = h_2 - h_1$: the heat transferred per kg equals the change in enthalpy of the stream. In a heat exchanger with two streams and no external heat loss, the enthalpy lost by the hot stream equals that gained by the cold stream: $\dot{m}_h(h_{h1} - h_{h2}) = \dot{m}_c(h_{c2} - h_{c1})$.

13.4 Throttling device

In a **throttling** process the fluid flows through a restriction – a partly open valve, an orifice or a porous plug – and suffers a large pressure drop. There is no work, the heat transfer is negligible, and although the velocity inside the restriction is high, the inlet and exit velocities in the pipe are small and nearly equal. The SFEE therefore gives $h_1 = h_2$: throttling is an **isenthalpic** process, and the enthalpy is the property that remains constant. Although the enthalpy is the same at the two ends, the process is highly irreversible (a large pressure drop with no useful effect). Throttling is used in refrigerator expansion valves, in the throttling calorimeter for measuring the dryness fraction of steam (Unit III), and in the Joule–Thomson experiment (Unit IV).

13.5 Worked examples on the SFEE

WORKED EXAMPLE 1.6 — Steam nozzle – mass flow, exit velocity and exit area

Problem. In a steam power plant, steam flows steadily from the boiler to the turbine through a horizontal nozzle of inlet area 0.08 m^2 . Steam at 1 MPa and 350°C enters at 60 m/s and leaves at 0.3 MPa and 250°C ; the nozzle loses 25 kW of heat to the surroundings. Determine (i) the mass flow rate, (ii) the exit velocity and (iii) the exit area. Take $v = 0.2825 \text{ m}^3/\text{kg}$ and $h = 3158.2 \text{ kJ/kg}$ at inlet, and $v = 0.7965 \text{ m}^3/\text{kg}$ and $h = 2967.9 \text{ kJ/kg}$ at exit.

Given: $A_1 = 0.08 \text{ m}^2$, $V_1 = 60 \text{ m/s}$, $v_1 = 0.2825 \text{ m}^3/\text{kg}$, $h_1 = 3158.2 \text{ kJ/kg}$; $v_2 = 0.7965 \text{ m}^3/\text{kg}$, $h_2 = 2967.9 \text{ kJ/kg}$; $\dot{Q} = -25 \text{ kW}$; horizontal, $w = 0$.

Formula: $\dot{m} = A_1V_1/v_1$; SFEE with $w = 0$, $\Delta z = 0$: $h_1 + V_1^2/2 + q = h_2 + V_2^2/2$; $A_2 = \dot{m}v_2/V_2$.

Solution.

(i) $\dot{m} = 0.08 \times 60/0.2825 = 16.99 \text{ kg/s}$.

Heat loss per kg: $q = -25/16.99 = -1.471 \text{ kJ/kg}$.

(ii) $V_2^2/2 = (h_1 - h_2) + V_1^2/2 + q = (3158.2 - 2967.9) + 60^2/2000 - 1.471 = 190.3 + 1.8 - 1.47 = 190.6 \text{ kJ/kg}$.

$V_2 = \sqrt{2 \times 190.6 \times 1000} = 617.4 \text{ m/s}$.

(iii) $A_2 = 16.99 \times 0.7965/617.4 = 0.0219 \text{ m}^2$.

ANSWER $\dot{m} = 17.0 \text{ kg/s}$; $V_2 \approx 617 \text{ m/s}$; $A_2 \approx 0.0219 \text{ m}^2$ (219 cm²). Note that the inlet kinetic energy (1.8 kJ/kg) and the heat loss (1.5 kJ/kg) are small compared with the enthalpy drop (190 kJ/kg).

WORKED EXAMPLE 1.7 — Air compressor – shaft work and pipe diameter ratio

Problem. Air flows steadily at 0.4 kg/s through a compressor, entering at 6 m/s, 100 kPa and 0.9 m³/kg and leaving at 4 m/s, 600 kPa and 0.2 m³/kg. The internal energy of the air leaving is 85 kJ/kg greater than that of the air entering, and cooling water in the jacket removes heat at 50 kW. Calculate (i) the rate of shaft work input and (ii) the ratio of the inlet pipe diameter to the outlet pipe diameter.

Given: $\dot{m} = 0.4 \text{ kg/s}$; $V_1 = 6 \text{ m/s}$, $p_1 = 100 \text{ kPa}$, $v_1 = 0.9 \text{ m}^3/\text{kg}$; $V_2 = 4 \text{ m/s}$, $p_2 = 600 \text{ kPa}$, $v_2 = 0.2 \text{ m}^3/\text{kg}$; $u_2 - u_1 = 85 \text{ kJ/kg}$; $\dot{Q} = -50 \text{ kW}$.

Formula: $q - w = (u_2 - u_1) + (p_2 v_2 - p_1 v_1) + (V_2^2 - V_1^2)/2$; continuity $A_1 V_1 / v_1 = A_2 V_2 / v_2$.

Solution.

Flow-work change: $p_2 v_2 - p_1 v_1 = 600 \times 0.2 - 100 \times 0.9 = 120 - 90 = 30 \text{ kJ/kg}$.

Enthalpy change: $h_2 - h_1 = 85 + 30 = 115 \text{ kJ/kg}$.

Kinetic energy change: $(4^2 - 6^2)/2000 = -0.01 \text{ kJ/kg}$ (negligible).

Heat per kg: $q = -50/0.4 = -125 \text{ kJ/kg}$.

$w = q - \Delta h - \Delta KE = -125 - 115 + 0.01 = -240.0 \text{ kJ/kg}$.

(i) Shaft work input $= 0.4 \times 240.0 = 96.0 \text{ kW}$.

(ii) $A_1/A_2 = (v_1/V_1)/(v_2/V_2) = (0.9/6)/(0.2/4) = 0.15/0.05 = 3.0 \Rightarrow d_1/d_2 = \sqrt{3} = 1.73$.

ANSWER Shaft power input = 96 kW; $d_1/d_2 = 1.73$ (the low-pressure inlet pipe is larger).

WORKED EXAMPLE 1.8 — Steady flow apparatus – change in internal energy

Problem. In a steady flow apparatus 130 kJ of work is done by each kg of fluid. At the inlet the specific volume, pressure and velocity are 0.35 m³/kg, 600 kPa and 15 m/s, and the inlet is 30 m above the floor. The discharge is at floor level with 0.6 m³/kg, 100 kPa and 250 m/s. The total heat loss is 8 kJ/kg. Does the specific internal energy increase or decrease, and by how much? Compare the sizes of the flow-work, kinetic-energy and potential-energy terms.

Given: $w = +130 \text{ kJ/kg}$; $q = -8 \text{ kJ/kg}$; inlet: $v_1 = 0.35$, $p_1 = 600 \text{ kPa}$, $V_1 = 15 \text{ m/s}$, $z_1 = 30 \text{ m}$; exit: $v_2 = 0.6$, $p_2 = 100 \text{ kPa}$, $V_2 = 250 \text{ m/s}$, $z_2 = 0$.

Formula: $u_2 - u_1 = (p_1 v_1 - p_2 v_2) + (V_1^2 - V_2^2)/2 + g(z_1 - z_2) + q - w$.

Solution.

Flow work: $p_1 v_1 = 210.0 \text{ kJ/kg}$; $p_2 v_2 = 60.0 \text{ kJ/kg}$; difference $= +150.0 \text{ kJ/kg}$.

Kinetic energy: $V_1^2/2 = 0.11 \text{ kJ/kg}$; $V_2^2/2 = 31.25 \text{ kJ/kg}$; difference $= -31.14 \text{ kJ/kg}$.

Potential energy: $g(z_1 - z_2) = 9.81 \times 30/1000 = +0.29 \text{ kJ/kg}$.

$u_2 - u_1 = 150.0 - 31.14 + 0.29 - 8 - 130 = -18.8 \text{ kJ/kg}$.

ANSWER The specific internal energy *decreases* by about 18.8 kJ/kg. The flow-work change (150 kJ/kg) and the work output (130 kJ/kg) dominate; the kinetic-energy term (31 kJ/kg) matters because of the high exit velocity; the potential-energy term (0.3 kJ/kg) is negligible even for a 30 m drop.

WORKED EXAMPLE 1.9 — Heat exchanger – turbine – nozzle plant

Problem. Air at 20 °C enters a heat exchanger at 25 m/s and is heated to 750 °C. It then enters a turbine at the same velocity and expands until its temperature falls to 600 °C, leaving the turbine at 50 m/s to enter a nozzle in which it expands until the temperature has fallen to 450 °C. The air flow rate is 1.5 kg/s. Taking $h = c_p t$ with $c_p = 1.005$ kJ/kg K, calculate (i) the rate of heat transfer to the air in the heat exchanger, (ii) the turbine power (no heat loss) and (iii) the nozzle exit velocity (no heat loss).

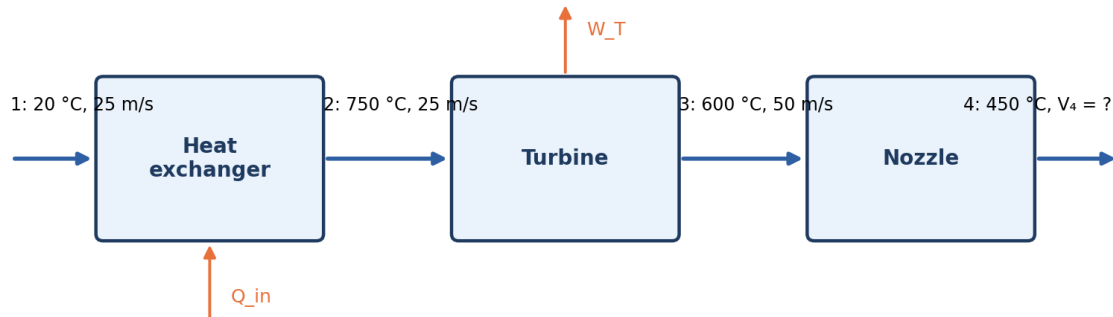


Figure 1.12 Layout of the plant in Worked Example 1.9.

Given: $\dot{m} = 1.5$ kg/s; $c_p = 1.005$ kJ/kg K; $t_1 = 20$, $t_2 = 750$, $t_3 = 600$, $t_4 = 450$ °C; $V_1 = V_2 = 25$ m/s, $V_3 = 50$ m/s.

Formula: Heat exchanger ($w = 0$, $\Delta KE = 0$): $\dot{Q} = \dot{m}c_p(t_2 - t_1)$. Turbine ($q = 0$): $\dot{W} = \dot{m}[c_p(t_2 - t_3) + (V_2^2 - V_3^2)/2]$. Nozzle ($q = 0$, $w = 0$): $V_4^2/2 = c_p(t_3 - t_4) + V_3^2/2$.

Solution.

(i) $\dot{Q} = 1.5 \times 1.005 \times (750 - 20) = 1100.5$ kW.

(ii) $\dot{W} = 1.5 \times [1.005 \times (750 - 600) + (25^2 - 50^2)/2000] = 1.5 \times [150.75 - 0.94] = 224.7$ kW.

(iii) $V_4^2/2 = 1.005 \times (600 - 450) \times 1000 + 50^2/2 = 150\,750 + 1250 = 152\,000$ J/kg $\Rightarrow V_4 = \sqrt{(2 \times 152\,000)} = 551$ m/s.

ANSWER $\dot{Q} = 1100$ kW; turbine power = 225 kW; nozzle exit velocity ≈ 551 m/s.

WORKED EXAMPLE 1.10 — Gas turbine – power, inlet pipe size and the effect of simplifying assumptions

Problem. Gas enters a turbine at 4 kg/s with a velocity of 60 m/s and an enthalpy of 950 kJ/kg, and leaves at 150 m/s with an enthalpy of 450 kJ/kg. Heat lost to the surroundings is 20 kJ/kg. For the gas $R = 0.287$ kJ/kg K; the inlet condition is 110 kPa and 30 °C. Determine (i) the power output and (ii) the inlet pipe diameter. Also find the percentage error in the power if both the kinetic-energy change and the heat loss are neglected, and comment on whether the simplification is acceptable.

Given: $\dot{m} = 4$ kg/s; $V_1 = 60$ m/s, $h_1 = 950$ kJ/kg; $V_2 = 150$ m/s, $h_2 = 450$ kJ/kg; $q = -20$ kJ/kg; $p_1 = 110$ kPa, $T_1 = 303.15$ K, $R = 0.287$ kJ/kg K.

Formula: $w = (h_1 - h_2) + (V_1^2 - V_2^2)/2 + q$; $\dot{W} = \dot{m}w$; $v_1 = RT_1/p_1$; $A_1 = \dot{m}v_1/V_1$; $d = \sqrt{(4A/\pi)}$.

Solution.

(i) $w = (950 - 450) + (60^2 - 150^2)/2000 - 20 = 500 - 9.45 - 20 = 470.55$ kJ/kg.

$\dot{W} = 4 \times 470.55 = 1882$ kW.

(ii) $v_1 = 0.287 \times 303.15/110 = 0.791$ m³/kg; $A_1 = 4 \times 0.791/60 = 0.0527$ m²; $d_1 = \sqrt{(4 \times 0.0527/\pi)} = 0.259$ m.

Simplified (ΔKE and q neglected): $w = 500$ kJ/kg, $\dot{W} = 2000$ kW; error = $(2000 - 1882)/1882 = 6.3$ %.

ANSWER Power = 1882 kW; inlet pipe diameter ≈ 0.26 m. Neglecting the kinetic-energy change and the heat loss over-estimates the power by about 6 %, which is too large for design or performance-guarantee purposes; the exit velocity of 150 m/s and the 20 kJ/kg heat loss each contribute a few per cent of the 500 kJ/kg enthalpy drop and must be retained. The simplification is acceptable only for a rough preliminary estimate.

WORKED EXAMPLE 1.11 — Reciprocating compressor – power, pipe areas and the energy budget

Problem. Air flows steadily at 0.5 kg/s through a reciprocating air compressor. It enters at 5 m/s, 1 bar and 0.8 m³/kg and leaves at 4 m/s, 6 bar and 0.17 m³/kg. The internal energy of the air leaving is 85 kJ/kg greater than that of the air entering, and the cooling water in the jacket absorbs heat at 55 kW. Determine (i) the power required to drive the compressor, (ii) the inlet and outlet pipe areas, and (iii) the share of the power input that goes into the internal-energy rise, the flow work and the heat rejected.

Given: $\dot{m} = 0.5$ kg/s; $V_1 = 5$ m/s, $p_1 = 100$ kPa, $v_1 = 0.8$ m³/kg; $V_2 = 4$ m/s, $p_2 = 600$ kPa, $v_2 = 0.17$ m³/kg; $u_2 - u_1 = 85$ kJ/kg; $\dot{Q} = -55$ kW.

Formula: work input = $(u_2 - u_1) + (p_2 v_2 - p_1 v_1) + (V_2^2 - V_1^2)/2 - q$; $A = \dot{m}v/V$.

Solution.

Flow-work change: $600 \times 0.17 - 100 \times 0.8 = 102 - 80 = 22$ kJ/kg. Kinetic-energy change: $(4^2 - 5^2)/2000 = -0.005$ kJ/kg (negligible). Heat rejected per kg: $55/0.5 = 110$ kJ/kg.

Work input per kg = $85 + 22 - 0.005 + 110 = 217.0$ kJ/kg.

(i) Power = $0.5 \times 217.0 = 108.5$ kW.

(ii) $A_1 = 0.5 \times 0.8/5 = 0.080$ m²; $A_2 = 0.5 \times 0.17/4 = 0.0213$ m².

(iii) Of the 217 kJ/kg input: internal-energy rise 85 kJ/kg (39 %), flow work 22 kJ/kg (10 %), heat rejected to the jacket water 110 kJ/kg (51 %).

ANSWER Power = 108.5 kW; $A_1 = 0.080$ m², $A_2 = 0.0213$ m²; the heat carried away by the cooling water is the largest single item – about half the shaft work – followed by the temperature rise of the air.

14. Unsteady flow processes – charging and discharging of vessels

When a tank is filled from a supply line or emptied through a valve, the mass and energy inside the control volume change with time: the flow is **unsteady** (transient). The general control-volume energy balance over the whole process, neglecting kinetic and potential energy, is

$$Q - W = (m_2 u_2 - m_1 u_1)_{cv} + \sum m_e h_e - \sum m_i h_i$$

where subscripts 1 and 2 refer to the initial and final contents of the tank, and i and e to the entering and leaving streams (each carrying its enthalpy, i.e. internal energy plus flow work).

Charging an evacuated insulated tank. Let a rigid, insulated, initially evacuated tank ($m_1 = 0$) be filled from a large supply line in which the fluid is at h_i . With $Q = 0$, $W = 0$ and no exit stream, $m_2 u_2 = m_i h_i$ and $m_2 = m_i$, so $u_2 = h_i$: the internal energy of the fluid in the tank equals the enthalpy in the line. The flow work done by the line fluid in pushing the mass into the tank appears as extra internal energy, so the tank fluid ends hotter than the line. For an ideal gas $c_v T_2 = c_p T_i$, giving $T_2 = \gamma T_i$ – for air from a line at 300 K the tank temperature rises to $1.4 \times 300 = 420$ K (147 °C).

Discharging a tank. If a tank is emptied slowly through a valve with no heat transfer, the fluid remaining in the tank undergoes a reversible adiabatic expansion (the leaving fluid takes its enthalpy with it), so its

pressure and temperature fall according to $pV^\gamma = C$. These results are used in the analysis of air receivers, gas-cylinder filling and pressure-vessel blow-down.

15. Practice problems (answers given)

No.	Problem	Answer
P1	A gas expands quasi-statically from 0.1 m ³ at 5 bar to 0.4 m ³ according to $pV^{1.25} = C$. Find the final pressure and the work done.	$p_2 = 0.884$ bar; $W = 58.6$ kJ
P2	2 kg of air at 300 K is heated at constant volume until its pressure doubles. Find ΔU and Q ($c_v = 0.718$ kJ/kg K).	$\Delta U = Q = 430.8$ kJ
P3	A closed system receives 90 kJ of heat and does 40 kJ of work in process 1–2; in process 2–1 it rejects 70 kJ of heat. Find the work of process 2–1 and ΔU_{1-2} .	$W_{2-1} = -20$ kJ; $\Delta U_{1-2} = 50$ kJ
P4	1 kg of air at 1 bar, 27 °C is compressed isothermally to 6 bar. Find W and Q ($R = 0.287$ kJ/kg K).	$W = Q = -154.3$ kJ
P5	Air at 1 bar and 300 K is compressed reversibly and adiabatically to 8 bar. Find the final temperature and the work per kg ($\gamma = 1.4$).	$T_2 = 543.4$ K; $W = -174.7$ kJ/kg
P6	Steam with $h = 3000$ kJ/kg enters a nozzle at 40 m/s and leaves with $h = 2750$ kJ/kg. Find the exit velocity (adiabatic).	708 m/s
P7	A turbine receives 3 kg/s of gas with $h_1 = 1000$ kJ/kg and delivers it with $h_2 = 620$ kJ/kg; heat loss 15 kW; velocities negligible. Find the power.	1125 kW
P8	A rigid insulated evacuated bottle is filled with helium ($\gamma = 1.67$) from a line at 290 K. Find the final temperature in the bottle.	484 K

16. Summary and formula sheet

- A **system** is what we choose to study; the **boundary** decides which heat, work and mass transfers enter the analysis. Closed systems have fixed mass; open systems (control volumes) admit mass; isolated systems admit neither mass nor energy.
- **Properties** are point functions (exact differentials, $\oint = 0$); **heat and work** are path functions – transient boundary phenomena that a system never "contains".
- A **quasi-static** process passes through equilibrium states and can be drawn on a property diagram; a **reversible** process is quasi-static and free of friction and other dissipative effects.
- **Zeroth law**: two bodies each in thermal equilibrium with a third are in thermal equilibrium with each other – the basis of temperature. T (K) = t (°C) + 273.15; the triple point of water is 273.16 K.
- **Displacement work** $W = \int p \, dV$ = area under the p – V curve (quasi-static only). Constant pressure: $p(V_2 - V_1)$; constant volume: 0; isothermal: $p_1 V_1 \ln(V_2/V_1)$; polytropic: $(p_1 V_1 - p_2 V_2)/(n - 1)$; adiabatic: same with $n = \gamma$.
- **First law**: $\oint \delta Q = \oint \delta W$ for a cycle; $Q - W = \Delta U$ for a non-flow process (Q positive in, W positive out). Energy is a property; the energy of an isolated system is constant; a PMM1 is impossible.
- **Enthalpy** $h = u + pv$; **specific heats** $c_v = (\partial u / \partial T)_v$, $c_p = (\partial h / \partial T)_p$; for an ideal gas $du = c_v dT$, $dh = c_p dT$ and $c_p - c_v = R$.
- **SFEE**: $h_1 + V_1^2/2 + gz_1 + q = h_2 + V_2^2/2 + gz_2 + w$, with continuity $\dot{m} = AV/v$. Nozzle/diffuser: $\Delta(V^2/2) = -\Delta h$; turbine: $w = h_1 - h_2$; compressor: $w_{in} = h_2 - h_1 + q_{rej}$; boiler/heat exchanger: $q = \Delta h$; throttling: $h_1 = h_2$.
- **Unsteady charging** of an evacuated insulated tank: $u_2 = h_{line}$, $T_2 = \gamma T_{line}$ for an ideal gas.

17. Review questions

17.1 Short-answer (two-mark) questions

1. What is a thermodynamic system? Name its three types.
2. Give one example each of a closed and an open system and state how they differ.
3. Classify the following as intensive or extensive: pressure, volume, density, total enthalpy.
4. What are the three conditions for a system to be in thermodynamic equilibrium?
5. Define a quasi-static process. Why is the concept useful?
6. How does a path function differ from a point function? Give an example of each.
7. What is a reversible process? Mention two factors that make a process irreversible.
8. State the Zeroth law and explain how it leads to the concept of temperature.
9. What property remains constant during throttling, and why?
10. Define c_v and c_p of a gas.
11. What is the triple point of water and what value is assigned to it on the Kelvin scale?
12. How are c_p , c_v and R related for a perfect gas?
13. Write the expression for the work done in a polytropic process and state when it cannot be used.
14. What is an isolated system? Give an example.
15. State the sign convention adopted for heat and work in this course.
16. How does a thermodynamic cycle differ from a process?
17. State the first law for a closed system executing a cycle.
18. Is internal energy a point function or a path function? Justify.
19. Define enthalpy. What is its expression for a perfect gas?
20. Mention any two limitations of the first law.
21. What is a PMM1? Why is it impossible?
22. Write the steady flow energy equation and name each term.
23. List the assumptions made in deriving the steady flow energy equation.
24. In what ways do heat and work differ as energy interactions?
25. How does a diffuser differ from a nozzle?
26. Define an adiabatic process.
27. Name four devices that are analysed with the steady flow energy equation.
28. Distinguish between heat and temperature.
29. Explain, using the SFEE, how a nozzle increases the velocity of a fluid.

17.2 Long-answer (thirteen-mark) questions

1. A closed system undergoes a cycle of four processes; the sum of the heat transfers per cycle is -200 kJ and the system completes 150 cycles per minute. Given $Q_{ab} = 0$, $W_{ab} = 2500$ kJ/min; $Q_{bc} = 23000$ kJ/min, $W_{bc} = 0$; $Q_{cd} = -2500$ kJ/min, $\Delta E_{cd} = -34000$ kJ/min, complete the table of Q , W and ΔE for all four processes and find the net rate of work in kW.
2. An ideal gas of volume 0.12 m³ at 14 bar and 520 K expands isothermally to five times its volume, is cooled at constant volume to 280 K and is then compressed polytropically to the initial state. Sketch the cycle and find the polytropic index, the net work and the net heat transfer.
3. Air of volume 0.3 m³ at 4.5 bar and 120 °C expands reversibly and adiabatically to 1.1 bar and is then heated at constant pressure until its enthalpy rises by 70 kJ. Find the total work and the index of a single polytropic process giving the same work between the same end states ($c_p = 1.005$, $c_v = 0.718$ kJ/kg K).
4. A cylinder holds 0.5 m³ of gas at 1.1 bar and 70 °C; it is compressed to 0.14 m³ and 5.5 bar. With $\gamma = 1.4$ and $R = 0.29$ kJ/kg K find the mass, the index of compression, the change in internal energy and the heat transferred, and state whether the process lies between the isothermal and adiabatic compressions.
5. A gas executes a cycle: 1–2 at constant pressure 1.6 bar from $V_1 = 0.025$ m³ with $W_{1-2} = 11$ kJ; 2–3 compression with $pV = C$ and $U_3 = U_2$; 3–1 at constant volume with $U_1 - U_3 = -25$ kJ. Sketch the cycle, find the net work and Q_{1-2} , and verify that $\sum Q = \sum W$.

6. Derive the steady flow energy equation from the first law, stating the assumptions, explaining the origin of the flow-work term and the meaning of each term, and giving the unit-mass and rate forms.
7. Apply the SFEE to a nozzle, a diffuser, a steam turbine and a rotary compressor; for each device sketch the arrangement, state the assumptions and obtain the reduced form of the equation.
8. Steam at 1 MPa, 300 °C ($v = 0.2579 \text{ m}^3/\text{kg}$, $h = 3051.2 \text{ kJ/kg}$) enters a horizontal nozzle of inlet area 0.12 m^2 at 80 m/s and leaves at 0.3 MPa, 200 °C ($v = 0.7163 \text{ m}^3/\text{kg}$, $h = 2865.6 \text{ kJ/kg}$); the heat loss is 20 kW. Find the mass flow, the exit velocity and the exit area.
9. Air at 0.6 kg/s enters a compressor at 8 m/s, 100 kPa, $0.92 \text{ m}^3/\text{kg}$ and leaves at 6 m/s, 650 kPa, $0.2 \text{ m}^3/\text{kg}$; the internal energy rises by 95 kJ/kg and 65 kW of heat is removed by the jacket water. Find the shaft power and the ratio of inlet to outlet pipe diameters.
10. In a steady-flow device 120 kJ/kg of work is done by the fluid; inlet $0.4 \text{ m}^3/\text{kg}$, 550 kPa, 20 m/s, 25 m above the exit; exit $0.65 \text{ m}^3/\text{kg}$, 100 kPa, 240 m/s; heat loss 10 kJ/kg. Determine the change in specific internal energy and compare the magnitudes of the terms.
11. Air at 25 °C enters a heat exchanger at 30 m/s and leaves at 700 °C; it then expands in a turbine to 560 °C, leaving at 55 m/s, and finally in a nozzle to 420 °C. For 2.2 kg/s and $c_p = 1.005 \text{ kJ/kg K}$ find the heat-exchanger duty, the turbine power and the nozzle exit velocity.
12. Gas enters a turbine at 6 kg/s, 45 m/s, $h = 920 \text{ kJ/kg}$ and leaves at 140 m/s, $h = 430 \text{ kJ/kg}$ with a heat loss of 22 kJ/kg; inlet 105 kPa, 25 °C, $R = 0.29 \text{ kJ/kg K}$. Find the power and inlet pipe diameter, and evaluate the error in the power if kinetic-energy change and heat loss are ignored.
13. Air at 0.45 kg/s enters a reciprocating compressor at 6 m/s, 1 bar, $0.85 \text{ m}^3/\text{kg}$ and leaves at 5 m/s, 6.5 bar, $0.16 \text{ m}^3/\text{kg}$; the internal energy rises by 88 kJ/kg and 50 kW is rejected to the jacket. Find the power input and the pipe areas, and compare the contributions of the internal-energy rise, the flow work and the heat rejected.

Text: P. K. Nag, "Engineering Thermodynamics", 6th Edn., McGraw-Hill — Chapters 1–5. References: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.; R. K. Rajput, "A Text Book of Engineering Thermodynamics", 5th Edn.